2 - The Nature of Econometrics and Economic Data

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Agenda

- What is Economic Data?
  - Cross-Sectional Data
  - Time Series Data
  - Panel (Longitudinal) Data
  - Observational, Experimental Data, and Survey Data
  - Challenges of Business Data
- Course Framing

What is Economic Data?

- In business analytics, we analyze **real-world data** to understand, forecast, and guide decisions.
- The most important **data structures** found in applied work are:
  - **Cross-sectional**: snapshot at one point in time.
  - **Time series**: measurements over time.
  - **Panel**: multiple observations over time for each unit.
- Different types of data require different econometric methods for analysis.

Cross-Sectional Data in Business Analytics

- **Cross sectional data** is collected at a single point in time across various units or subjects (e.g., individuals, firms, cities).
- **Implicit assumption:** all observations are independent, and the sample represents a snapshot in time.

```r
## Load dataset
cross_data <- read_csv("data/cross_data.csv")

## See first few rows of the data
head(cross_data, 5)
```

```
# A tibble: 5 x 5
  EmployeeID Satisfaction Salary Department Tenure
       <dbl>        <dbl>  <dbl> <chr>       <dbl>
1          1         2.50 50870. HR              3
2          2         4.80 47010. Sales           7
3          3         3.93 50918. HR              2
4          4         3.39 30124. Sales           2
5          5         1.62 47803. Finance         7
```

Analysis of Cross-Sectional Data (1/2)

Cross sectional data is commonly used for descriptive **snapshot analyses**, e.g., average satisfaction level.

```
# Summary statistics
stargazer(data.frame(cross_data |> select(-EmployeeID)), type = "latex", digits = 2
        , title = "Summary Statistics of Cross-Sectional Data", header = FALSE)
```

Table 1: Summary Statistics of Cross-Sectional Data

Statistic	N	Mean	St. Dev.	Min	Max
Satisfaction	100	2.88	1.19	1.02	4.95
Salary	100	49,989.20	9,129.29	30,124.31	74,632.42
Tenure	100	5.72	2.92	1	10

Analysis of Cross-Sectional Data (2/2)

Cross sectional data is commonly used for descriptive **snapshot analyses**, e.g., how satisfaction relates to salary.

```
# Correlation table
correlation_matrix <- cor(cross_data |> select(Satisfaction, Salary, Tenure))

# Format output
kable(round(correlation_matrix, 2),
      caption = "Correlation Matrix: Cross-Sectional Data")
```

Table 2: Correlation Matrix: Cross-Sectional Data

	Satisfaction	Salary	Tenure
Satisfaction	1.00	-0.15	-0.03
Salary	-0.15	1.00	0.07
Tenure	-0.03	0.07	1.00

Time Series Data in Business Analytics

- In **time series data** observations on one or more variables are collected over time and ordered chronologically. Examples include stock prices, consumer price index, GDP, sales figures, etc.
- **Implicit assumption:** observations are dependent over time, and there are patterns or trends that can be modeled.

```
## Load dataset
time_series_data <- read_csv("data/time_series_data.csv")

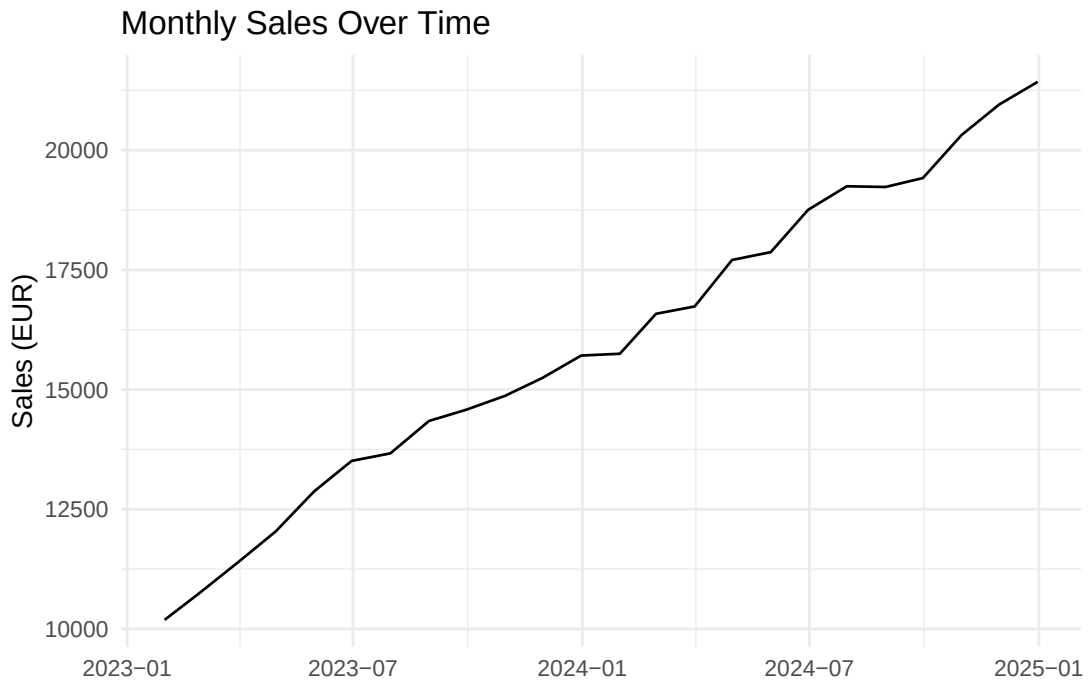
## See first few rows of the data
head(time_series_data, 5)
```

```
# A tibble: 5 x 3
  Month       Sales AdSpend
  <date>      <dbl>   <dbl>
1 2023-01-31 10190.   5288.
2 2023-02-28 10751.   4541.
3 2023-03-31 11395.   5411.
4 2023-04-30 12035.   5456.
5 2023-05-31 12877.   6390.
```

Analysis of Time Series Data

Time series data is commonly used for **forecasting**, trend analysis, and seasonality detection.

```
# Plot time series
ggplot(time_series_data, aes(x = Month, y = Sales)) + geom_line() +
  labs(title = "Monthly Sales Over Time", y = "Sales (EUR)", x = "") +
  theme_minimal()
```



Panel (Longitudinal) Data in Business Analytics (1/2)

- **Panel datasets** contain observations on the *same* cross-sectional units (e.g., individuals, firms, products) over time.
- This type of data combines benefits of both cross-sectional and time series and allows for controlling individual-specific characteristics that do not change over time (*unobserved heterogeneity*).
- In organizing panel data we order the data by **unit** of analysis and then **period**.

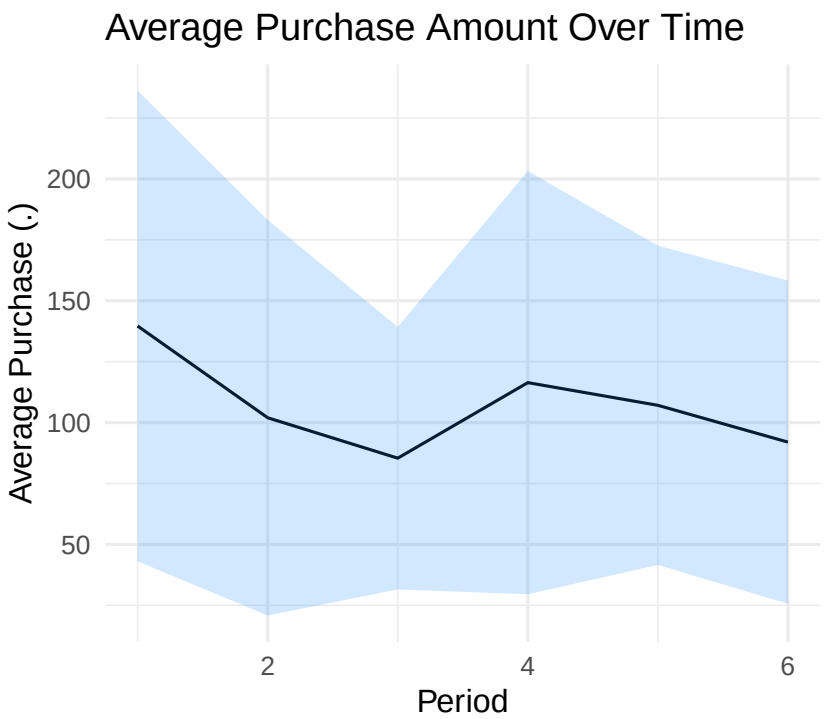
Panel (Longitudinal) Data in Business Analytics (2/2)

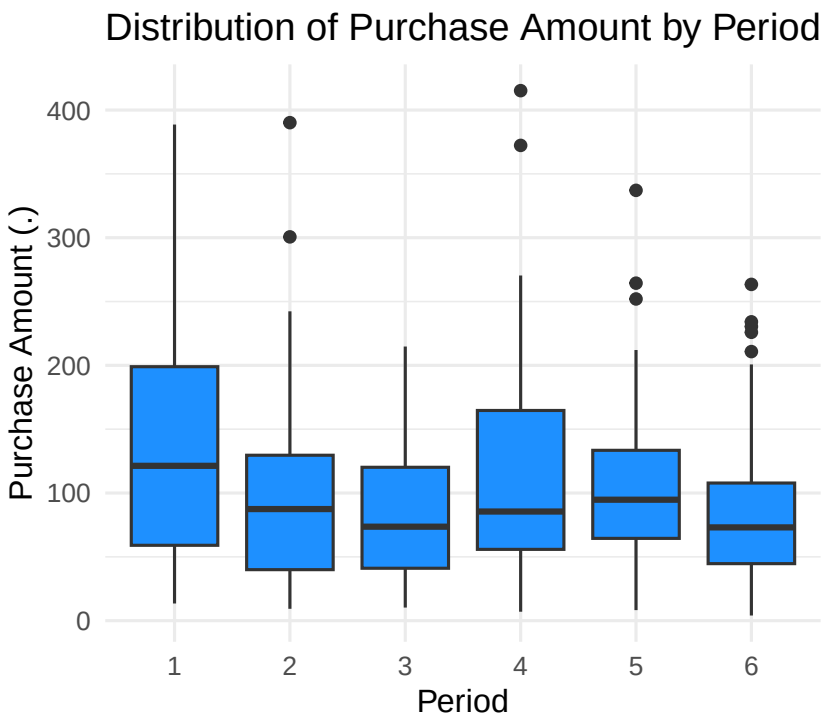
```r
## Load dataset
panel_data <- read_csv("data/panel_data.csv")

## See first few rows of the data
head(panel_data, 15)
```

```
# A tibble: 15 x 5
   CustomerID Period PurchaseAmount EmailCampaign LoyaltyScore
        <dbl>  <dbl>          <dbl>         <dbl>        <dbl>
 1          1      1           90.2             1         86.4
 2          1      2          102.              1         85.0
 3          1      3          179.              0         73.7
 4          1      4           63.1             0         54.2
 5          1      5          100.              1         64.3
 6          1      6          106.              0         65.4
 7          2      1          155.              0         56.7
 8          2      2          120.              0         81.5
 9          2      3           20.7             0         55.5
10          2      4           60.9             0         89.7
11          2      5           86.2             1         89.6
12          2      6           28.1             1         66.1
13          3      1          103.              0         73.3
14          3      2           71.8             0         69.3
15          3      3           73.2             0         79.0
```

Panel data is commonly used for **dynamic analyses** that account for both individual and time effects.





Observational vs Experimental Data

Most business data is **observational**:

- Collected passively during operations (e.g., sales logs, customer transactions).
- No control over variable assignment by the analyst.

Experimental data:

- Researcher **controls treatment assignment** (e.g., A/B tests in marketing).
- Stronger internal validity for identifying **causal effects**.

Survey Data

Survey data often falls under observational:

- Based on **self-reported** responses.
- Researchers typically **do not assign treatments** — they ask about experiences, attitudes, or past behaviors.
- But! Surveys can **embed experiments**:
  - Example: Varying wording or framing of questions to test response sensitivity.

Challenges of Self-Reported (Survey) Data

Survey data is **common** in business (e.g., customer satisfaction, employee engagement, brand perception), but it comes with **unique challenges**:

- **Sampling bias**: Who chooses to respond? Are they representative?
- **Response bias**: Are people answering truthfully or strategically?
- **Measurement error**: Are we capturing the construct we care about?

Takeaway: Even when data is "clean", understanding its **origin** and **limitations** is crucial for credible analysis.

Why Does the Distinction Matter?

The type of data affects **what you can claim**:

- **Experimental data**: best for identifying **causal relationships**.
- **Observational data**: useful for **correlations** and **associations**, but **causality requires strong assumptions**.

How can we trust that the observed effect is due to the variable of interest and not something else?

- Econometrics helps us address this by:
  - Providing tools to account for potential confounding factors.
  - Offering a range of methods to isolate causal effects from observational data.

The Reality of Data Quality

"Problems in Business Intelligence are mostly data quality related - the visualization part is the reward after the tough work." *Oskar Tumnitz, Class of MSc Business Analytics 2024*

Business data is often:

- **Messy** (missing values, duplicates, inconsistent formats)
- **Noisy** (errors, outliers, manual entry)
- **Biased** (non-representative samples, selection issues)

Careful data **collection, cleaning and validation** are **essential** steps in the analytics process and often where most time is spent!

Recommended readings

- Wooldridge, *Introductory Econometrics*, Chapter 1